MODEL IDENTIFICATION AND DATA ANALYSIS – Part 1 (Automation&Control Engineering), A.Y. 2021/2022

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Surname	**Name**	**University ID Number**	**Signature**
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- Write the solutions in the <u>blank</u> areas (use the back of the page, if needed)
- The number of pages is 4. Additional papers <u>will not be considered</u>.
- <u>Clarity, order and precision</u> are strongly considered for the final evaluation.

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1. [Neural networks]

a. Explain what a Feed-forward Neural Network (FF-NN) is and draw the scheme of a 2-layer FF-NN.

b. What would happen to the input/output mathematical relation if the activation function was simply the weighted sum of all input signals?

c. Explain why the simple activation scheme described in the previous point is not normally used in FF-NNs, although it would simplify and accelerate the calculations for the backpropagation algorithm.

Solutions

a. See the lecture notes (chapter 09).

b. In this situation, the output of the network could also be generated by a single-layer network. In fact, if weighted sums of the input signals are fed into a neuron at the next level, that neuron will still just compute a weighted sum of the outputs of the neurons from the previous layer, which is a weighted sum of the input signals to the network.

To be more rigorous, we can write the weights from the input to the hidden layer as a matrix W^{HI}, the weights from the hidden to output layer as W^{OH}, and the bias at the hidden and output layer as the vectors b^{H} and b^{O}. Using vector and matrix multiplication, the hidden activations H can be written as a function of the inputs I as

$$H = b^{H} + W^{HI} * I$$

The output activations can then be written as

$$O = b^{O} + W^{OH} * H = b^{O} + W^{OH} * (b^{H} + W^{HI} * I) = (b^{O} + W^{OH} * bH) + (W^{OH} * W^{HI}) * I$$

Because of the associativity of matrix multiplication, this can be written as

$$O = b^{OI} + W^{OI} * I$$

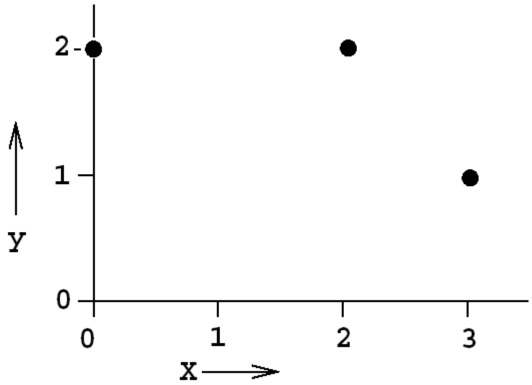
with obvious definitions for b^{OI} and W^{OI}. Therefore, the same hypothesis set can be described with a simpler network, with no hidden layers, using the bias b^{OI} and weights W^{OI}.

c. The resulting model would be a simple linear regression model, with limited accuracy for complex data generation mechanisms. Indeed, computation of the solution via gradient descent (backpropagation) would be much faster and robust with respect to the initialization. An analytical solution could be finally retrieved. See the lecture notes (chapter 09) for further details.

2. [Linear regression]

Suppose you have the dataset below with one real-valued input and one real-valued output.

x	y
0	2
2	2
3	1



a. What is the mean squared leave-one-out cross validation error of using linear regression?
b. Suppose we use the trivial constant model $y = c$. What is the mean squared leave-one-out error in this case? (Assume c is learned from the non-left-out data points).
c. Describe how L_2 regularization works and explain why it is a good approach to combat overfitting when the number of data is small (like in this case).

Solutions

a. The mean squared leave-one-out cross validation error is

$$\frac{2^2 + \left(\frac{2}{3}\right)^2 + 1^2}{3} = \frac{49}{27}$$

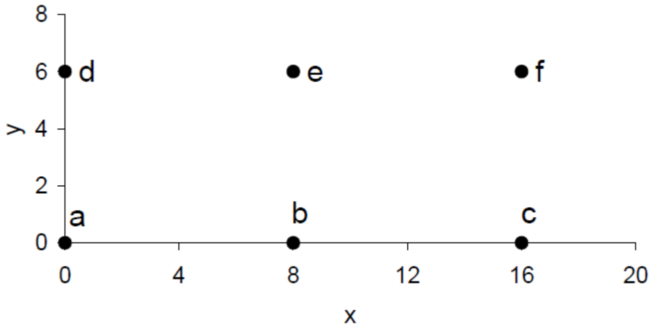
b. The answer changes as follows

$$\frac{0.5^2 + 0.5^2 + 1^2}{3} = \frac{1}{2}$$

c. See the lecture notes (chapter 08).

3. [k-means]

Consider a set S of 6 points in the plane as shown in the figure below.



We run the k-means algorithm on these points with $k = 3$. The algorithm uses the *Euclidean* distance metric (i.e., the straight-line distance between two points) to assign each point to its nearest centroid. Two definitions:

• A *k-starting configuration* is a subset of k starting points from S that form the initial centroids, e.g., {a, b, c} is a 3-starting configuration.

• A *k-partition* is a partition of S into k non-empty subsets, e.g., {a, b, e}, {c, d}, {f} is a 3-partition.

Clearly any k-partition induces a set of k centroids. A k-partition is called *stable* if a repetition of the k-means iteration with the induced centroids leaves it unchanged.

a. How many 3-starting configurations are there?

b. Fill in the following table

3-partition	Stable?	An example of *3-starting configuration* that can arrive at the 3-partition after 0 or more iterations (write "none" if it does not exist)
{a, b, e}, {c, d}, {f}		
{a, b}, {d, e}, {c, f}		
{a}, {d}, {b, c, e, f}		
{a, b, d}, {c}, {e, f}		

c. Describe the Lloyd's algorithm for clustering via K-means.

Solutions

a. There are $\binom{6}{3} = 20$ 3-starting configurations (in fact, there are $\binom{n}{k}$ ways to choose an unordered subset of k elements from a fixed set of n elements).

b. The table could be filled as follows.

3-partition	Stable?	An example of *3-starting configuration* that can arrive at the 3-partition after 0 or more iterations (write "none" if it does not exist)
{a, b, e}, {c, d}, {f}	N	none
{a, b}, {d, e}, {c, f}	Y	{b, c, e}
{a}, {d}, {b, c, e, f}	Y	{a, b, d}
{a, b, d}, {c}, {e, f}	Y	{a, c, f}

c. See the lecture notes (chapter 10).

4. [Learning theory]

Explain qualitatively (by using the learning curves) why the hypothesis set should depend on the available dataset. Why the previous statement might be true even in the (ideal) case where the complexity of the target function is known?

Solutions

a. See the lecture notes (chapter 08).